TEAM AVRON

CRITERIA OF SUCCESS

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NEED STATEMENT

"Visually impaired individuals navigating low-resource outdoor environments need real-time detection of environmental hazards to support independent mobility."

CRITERIA OF SUCCESS

1. Obstacle detection reliability (≥ 90%)

Detects static obstacles from knee to head height beyond white cane reach, with at least 90% obstacle detection reliability in outdoor conditions, tested across varied obstacle sizes and materials.

2. Range of detection (0.3 m – 3 m)

Senses obstacles from 0.3 m to 3 m ahead. Short-range mode covers 0.3 m to 1.5 m and long-range mode extends to 3 m, with distance accuracy verified at set points.

3. Real-time response time (300 ms)

Alert latency target of 300 ms from detection to user feedback. At a walking speed of about 1.2 m/s the user moves only about 0.36 m during this delay, leaving enough time to react.

4. Documented safety and risk control (ISO 14971)

Maintains an ISO 14971-style risk management file with no unacceptable open risks. Required safety features include ; low-voltage battery power, biocompatible, skin-safe contact materials and low-battery alerts before shutdown

5. Field of detection (≥ 120°)

Covers a horizontal arc of at least 120° in front of the user, so obstacles to the sides are detected without precise cane sweeping.

6. Battery life (≥ 8 hours)

Runs for at least 8 hours of continuous use on one charge, verified by a runtime test until the low-battery alert, which must trigger before shutdown.

7. Usability (SUS ≥ 70)

System Usability Scale score of at least 70. When 7 to 10 people test the device, at least 7 must succeed in mounting, detaching, powering on and switching modes unaided using tactile cues, with zero critical use errors.

8. Weather resistance (IP54)

Target of IP54 protection against dust and water splashes, with reliable operation in outdoor noise and daylight (minimum acceptable level IPX2).

9. Distinct alerts by hazard type

Gives distinct vibration or sound patterns that differentiate; Ground hazards (drop-offs, steps, puddles), trunk and head-level hazards, and distance bands (near, mid and far)

10. Moving hazard alerting

Detects fast-approaching dynamic objects such as vehicles or pedestrians and gives a priority alert in time for the user to react safely.